T10 - Trauma Arrest
Effective: January 21, 2021

**DO NOT ATTEMPT RESUSCITATION** If the patient is

- Unresponsive
- Apneic
- Has non reactive pupils
- No palpable pulse AND signs of obvious death: lividity, rigor mortis or venous pooling
  OR burned beyond recognition, decapitation, hemicorpectomy, or decomposition.

**Initially utilize manual C-spine immobilization and apply cervical collar if situation allows. Patient may remain on scoop or LBB if used.**

**If MOI appears inconsistent with traumatic cardiac arrest** (i.e. Ground level falls with minimal external trauma, minor NVC, etc.) consider medical cause of the arrest and treat in accordance with Medical ACLS and Medical Arrest protocols.

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| EMS witnessed arrest of patent with *penetrating* trauma | | Load and Go<br>Trauma Arrest care in route | |
| Arrest in pregnant female ≥ 20 weeks gestation or uterine fundus is palpable at or above the umbilicus | | Load and Go<br>Trauma arrest care in route<br>Transport with uterus displaced to left | |
| Child ≤ 14 | | Follow Pediatric Trauma Arrest care | Do not discontinue resuscitation in the field |
| Hypothermia | | Load and Go<br>Refer to hypothermia protocol | Do not discontinue resuscitation |
| Drowning | Refer to Submersion protocol.<br>Stay on scene for 20 minutes and deliver Medical ACLS care in accordance with rhythm | | Do not discontinue resuscitation in the field |
| Lightning strike/Electrocution | Stay on scene for 20 minutes and deliver Medical ACLS care in accordance with rhythm | | Do not discontinue resuscitation in the field. |

Spinal Motion Restriction may be initially maintained by applying manual immobilization in a neutral position. If transport is initiated, apply cervical collar and secure torso and head.

Once arrest is recognized begin chest compressions in accordance with AHA standards. Do not use LUCAS device while on scene. The LUCAS may be used only during transport.

## Trauma Arrest Resuscitation:

- Immediately control any life threatening hemorrhage in accordance with Multi-system Trauma Protocol
- Initiate CPR in accordance with AHA standards.
- If BVM is adequate (good air entry/ chest rise), continue with trauma arrest protocol**. If not,** an ETT or I gel should be placed and confirmed with capnography.
- Perform bilateral needle chest decompressions **unless** there is a hanging, isolated head or isolated extremity injuries Bilateral simple finger thoracostomies may be performed in lieu of needle chest decompressions.
- *Begin infusion of 1 liter of normal saline under pressure.*

***Goal for completion of above care should not exceed 5 minutes.***

**AFTER ABOVE ARE COMPLETE:**

***If pulse palpable:*** immediately Load and go. Follow ROSC protocol. If arrest reoccurs during transport continue trauma arrest as noted below. **Do not discontinue efforts once in route.**

***If pulse not palpable:*** Apply cardiac monitor and treat based on rhythm per chart

**PLEASE NOTE EXCEPTION BELOW for Trauma due to hanging or isolated head injury.**

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| **Trauma Arrest VF/PVT** | • VF/pulseless VT, defibrillate as needed at each 2 minute rhythm check.<br>• Repeat fluid boluses regardless of crackles.<br>• Place advanced airway with continuous ETCO2 waveform monitoring<br>• Do not administer any medications<br>• Treat any identified reversible causes. | Load and Go now | L |
| **Trauma Arrest Organized Rhythm > 40 bpm** | • Repeat fluid boluses regardless of crackles.<br>• Place advanced airway with continuous ETCO2 waveform monitoring<br>• Do not administer any medications<br>• Treat any identified reversible causes. | Load and go now | L |
| **Organized rhythm ≤ 40 BPM PEA OR Asystole upon monitor check**<br>**Consider Termination** | • Ensure a minimum of 500 ml of NS has been infused.<br>• Contact Medical Control to confirm discontinuation of resuscitation<br>• Terminate resuscitation (in field only)<br>• Document contact with Medical Control and name of physician | Load and Go only if advised by Medical Control<br>Treat as outlined without administration of medications. | |

| **Organized rhythm ≤ 40 BPM PEA OR Asystole upon monitor check** **<u>Consider Termination</u>** | • Call coroner and law enforcement<br>• Leave all airways, IV's, IO's, and electrodes in place.<br>• Paramedic is to stay on scene unless otherwise directed by BC until coroner arrives.<br>• If transport advised: Load and go |
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**EXCEPTION: Trauma arrest due to hanging or isolated head injury while on scene:**

- Prep and administer epinephrine 1 mg (1:10,000) IVP after the second shock in the case of VF/PVT and as soon as possible in asystole or PEA of any rate
- Repeat epinephrine in 4 minutes for a total dose of 2.
- At the conclusion of the next 4 minute cycle observe the monitor and follow criteria for termination if met or Load and Go treating as outlined without further administration of medication.

**Children ≤ 14 Trauma Arrest Resuscitation**

- Immediately control any life threatening hemorrhage in accordance with Multi-system Trauma Protocol
- Initiate CPR in accordance with AHA standards.
- If BVM is adequate, continue with trauma arrest protocol**. If not,** an ETT or I gel should be placed and confirmed with capnography.
- Perform bilateral needle chest decompressions **unless** there is a hanging, isolated head or isolated extremity injuries Child ≤ 20 kg use 18 gauge 2.5 inch decompression needle
  Child > 20 Kg use 14 gauge 3.25 inch decompression needle
- Begin infusion of 20 ml/Kg of normal saline under pressure.
- Place cardiac monitor defibrillate if needed.
- Transport now and follow Adult Trauma Arrest Care as above.
- In the case MOI is Hanging or Isolated Head injury, the 2 doses of epinephrine may be administered in the ambulance en route. Dose of epinephrine in accordance with Pediatric arrest protocol.
- ***<u>DO NOT DISCONTINUE RESUSCITATION IN THE FIELD REGARDLESS OF RHYTHM.</u>***